

Wang "Melvin" Ming

MSc in Information Engineering

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🌐 LinkedIn Profile

EDUCATION

- **The Chinese University of Hong Kong, Hong Kong** 2025-2026
Master of Science in Information Engineering Ongoing
- **University of Cambridge** 2022
Mathematics for Engineers Programme Certificate Granted
- **The Hong Kong Polytechnic University, Hong Kong** 2020-2024
Bachelor of Science in Computer Science Graduated with Second Class Honours Division I

EXPERIENCE

- **Department of Electrical and Computer Engineering, University of Waterloo** January 2026 - present
Co-researcher Canada
 - Exploring the cost structure of MLLM educational deployment in block-based programming.
- **Department of Information Engineering, CUHK** September 2025 - present
Research Project Hong Kong
 - MLLM Reasoning for Next Generation live-streaming Digital Human
 - QWen3-Omini + Wan-S2V Pipeline Construction, Baseline Deployment
- **ARISE Lab, Department of Computer Science and Engineering, CUHK** March 2025 - present
Co-researcher Hong Kong
 - Front-end Coding intelligence analysis for Multimodal LLMs Baseline
 - Data Collection and Processing, Metrics, Evaluation, Benchmarking
 - Building AI Agent system as Metareviewer for an academic paper review
 - Agentic AI, MLLM, RAG, MoE
- **Department of Computing, PolyU** June 2024 - September 2024
Research Assistant (Full-time) Hong Kong
 - Temporal Graph Neural Network (TGNN), Link Prediction and Node Classification
 - Model Testing, Evaluation Baseline testing and Benchmarking
- **Code for Asia Ltd.** June-August, 2023
Student internship Singapore
 - Front-end Development
 - AI Problem-solving Architect
- **Department of Building and Real Estate, PolyU** April 2023- June 2024
Research Assistant (Part-time) Hong Kong
 - LLM-driven Online Social Network Analysis
 - Data Retrieval, Large Language Model, Social Network Analysis, Paper Completion

PUBLICATIONS & PREPRINTS

- J. Xiao, M. Wang, M. H. Lam, Y. Wan, J. Liu, Y. Huo and M. R. Lyu, "DesignBench: A Comprehensive Benchmark for MLLM-based Front-end Code Generation", in arXiv preprint, 2025, doi: 10.48550/arXiv.2506.06251
Keywords: MLLM, Code Intelligence, Software Engineering, Benchmark
- M. Wang, R. Ma, G. Q. Shen and J. Xue, "How Large Language Model Empowers the Analysis of Online Public Engagement for Mega Infrastructure Projects: cases in Hong Kong", in IEEE Transactions in Engineering Management (IEEE-TEM), 2025, doi: 10.1109/TEM.2025.3553595
Keywords: Large Language Model, Social Network Analysis, Public Engagement, Mega Projects
- R. Ma, G. Q. Shen, P. Lou and M. Wang, "Large Language Model-based data-driven framework for digital transformation in construction industry," in ASCE International Conference on Computing in Civil Engineering 2024 (i3CE), doi: 10.1061/9780784486115.012
Keywords: Large Language Model, Digital Transformation, Engineering Management

PROJECTS

•Data vs Pipeline Parallelism in LLM Training

October 2025

Aims to build a VR Metaverse system for elderly rehabilitation training.

- Implements distributed training methods, including data parallelism and pipeline parallelism across multiple GPUs.

•Robotic-guided Self-help Rehabilitation in Metaverse for stroke patients

March - May 2023

Aims to build a VR Metaverse system for elderly rehabilitation training.

- Tools & technologies used: Leap Motion, Unity, UART for data transmission
- The forearm wearable device is developed by PolyU BRE

•LLM-empowered Family Healthcare Management System

January - March 2024

A smarter family healthcare system with personalized medical inquiry and app operation via word inputs.

- Tools & technologies used: Python, Java, LLM, Prompt Engineering

TECHNICAL SKILLS AND INTERESTS

Languages: English, Mandarin, Cantonese

Developer Tools: PyCharm, VSCode, IntelliJ IDEA

Programming Languages: C/C++, Java, Javascript, Python, R

Libraries: Torch, Tensorflow, NetworkX, Sklearn, Pandas, Matplotlib, Seaborn

Operating Systems: Windows, Linux

Areas of Interest: MLLM, Software Engineering, Agentic AI, Social Network Analysis

POSITIONS OF RESPONSIBILITY

•Reviewer, IEEE Transactions in Engineering Management (IEEE-TEM)

May 2025-October 2025

•Student Ambassador, Amazon Web Services(AWS)

2023-2024

•First Aid Brigade Member, Hong Kong St. John Ambulance

July 2024-Present

ACHIEVEMENTS

•Certificate of Outstanding Contribution, AWS Student Ambassador Programme

2023

•Excellent Teaching Team, STEM Education for Children, PolyU Service Learning Programme

2022

•3rd Prize, National Olympics for Informatics in Provinces (NOIP), China

2018